Essential in NX CAM (40 Hours)

S.No	Session	Topics	Type	Duration (Hours)
1	Session 1	**INTRODUCTION TO MANUFACTURING & NX-CAM** Introduction to manufacturing About manufacturing types About machining types Milling operations overview Introduction to CAM Benefits of CAM Introduction to NX CAM Benefits of NX CAM Milling overview About Conventional milling and Climb milling Milling machines overview Milling cutters overview **Manufacturing in NX** Understanding CAM terminologies Understanding CAM set up About CAM configuration About NC assistant About Milling processors and operation sub types	Theory/Demo	2.0Hrs
2	Session 2	Exercises included for Practice	Practicals	2.0Hrs
3	Session 3	**NX-CAM TOOLS** **Operation Navigator:** Operation Navigator overview, Operation status symbols Editing Operations, Transform Operations **Geometry Groups:** Geometry Groups overview, MCS overview In Process Work piece, Assigning material to part **Tool Groups:** About milling tool types, About tool holder Retrieving Tools from Tool Library **Program and Method Groups:** About Program groups, About Method group creation Defining Feed rates **Visualize and Analyze:** Understanding Tool Path Visualization, Understanding Visualization Replay Understanding Dynamic Visualization, Understanding IPW thickness by color About Gouge Checking, About Generating Tool Paths Parallel Generate overview, Verify Tool Path overview About Installation of Postprocessor, Generating program using Post process Generating Shop Documentation	Theory/Demo	2.0Hrs
4	Session 4	Exercises included for Practice	Practicals	2.0Hrs
5	Session 5	**MILLING OPERATION - I** **Facing Milling Operation:** Introduction to Face Milling operation, Specifying Check bodies Specifying face boundaries, Defining Path Settings, Defining Stopover Defining Cut Patterns Creating Face Milling Area Operation, Defining Cut Walls Defining Mixed and Manual Cut, Patterns Post process a program, Generating Shop Documentation **Planar Milling Operation:** Introduction to Planar milling operation Specifying part geometries, Stock overview Defining Non cutting moves, Defining Transfer/Rapid	Theory/Demo	2.0Hrs
6	Session 6	Exercises included for Practice	Practicals	2.0Hrs
7	Session 7	**MILLING OPERATION - II** **Planar Profile Operation** Creating Planar Profile Operation **Planar Milling Operation** Planar Operation using Standard Drive cut pattern **Planar Text Operation** Creating Planar Text Operation **Thread Milling Operation**	Theory/Demo	2.0Hrs

		Creating Thread Milling Operation on Hole and Boss feature		
		Contour Milling Operation		
		Creating Cavity Milling Operation		
		Defining Wave Geometry Linker		
		Defining Trochoidal cut pattern		
		Machining based on level based IPW		
8	Session 8	**Exercises included for Practice**	Practicals	2.0Hrs
9	Session 9	**MILLING OPERATION - III**	Theory/Demo	2.0Hrs
		Rest Milling Operation		
		Creating Rest Milling Operation		
		Defining Cut levels		
		Z-level Milling Operation		
		Creating Z Level Milling Operation		
		Defining Containment for blank		
		Creating Additional Passes		
		Defining Step over in Path Settings		
		Surface Contouring Operation		
		Creating Contour Area Non Steep Operation		
		Milling Operation based on Area Drive method		
		Creating Contour Area Operation		
		Defining Cut Direction		
		Editing Tool display		
		Defining Trim Boundary		
		Specifying Cut Angles		
		Specifying Directional Steep		
10	Session 10	Exercises included for Practice	Practicals	2.0Hrs
11	Session 11	**RADIAL CUT OPERATION**	Theory/Demo	2.0Hrs
		Creating Fixed Contour Operation		
		Defining Radial Drive method		
		Editing Band Parameter		
		Engraving Text on Contour Surface		
		Creating Contour Text Operation		
12	Session 12	Exercises included for Practice	Practicals	2.0Hrs
13	Session 13	**VARIABLE CONTOUR OPERATION**	Theory/Demo	2.0Hrs
		Creating Variable Contour Operation		
		Defining Normal to Drive method		
		Creating Variable Contour Operation using Normal to Part Tool Axis method		
		Defining Projection Vector and Tool Axis		
		Defining Lead Angle		
		Creating Variable Contour Operation using Interpolated Tool Axis vectors		
		Interpolating Tool Axis		
14	Session 14	Exercises included for Practice	Practicals	2.0Hrs
15	Session 15	**MULTI BLADE MILLING OPERATION**	Theory/Demo	2.0Hrs
		About Turbo machinery Milling Operation		
		Creating Multi Blade Roughing Operation		
		Specifying Hub geometry		
		Specifying Shroud geometry		
		Specifying Blade geometry		
		Specifying Blend geometry		
		Defining Drive Methods		
		Defining Cut Levels		
		Creating Multi Blade Hub Finish Operation		
		Creating Multi Blade Finish Operation		
16	Session 16	Exercises included for Practice	Practicals	2.0Hrs
17	Session 17	**DRILLING OPERATION**	Theory/Demo	2.0Hrs
		About Drilling machines		
		About Drilling tools		
		About Drilling tool material		
		Various Drilling operations		
		Operation sub types in Drilling Operation		
		Defining Drill geometry		
		Creating Spot Drilling Operation		
		Creating Drilling Operation		
		Creating Counter Boring Operation		

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		Creating Reaming Operation		
		Creating Peck Drilling Operation		
		Creating Boring Operation		
		Creating Counter Sinking Operation		
18	Session 18	Exercises included for Practice	Practicals	2.0Hrs
19	Session 19	**TURNING OPERATION**	Theory/Demo	2.0Hrs
		About Lathe machines		
		Various tools available for Turning Operation		
		Operation subtypes in Turning Operation		
		About Part geometry		
		About Blank geometry		
		About Avoidance		
		About Cut Region Containment		
		Defining Turning Work piece		
		Defining Trim Planes		
		Create Roughing and Facing Operation		
		Create Grooving and Boring operation		
		IPW definition in 2D and 3D		
		Create Spot and Peck Drilling Operation		
		Turning Operation using multiple spindles		
		Creating Teach Mode Operation		
		Creating Thread cutting in Turning Operation		
20	Session 20	Exercises included for Practice	Practicals	2.0Hrs